

CSS Backgrounds

The CSS background properties are used to add background effects for elements.

CSS background-color

The `background-color` property specifies the background color of an element.

Example

The background color of a page is set like this:

```
body {  
  background-color: lightblue;  
}
```

With CSS, a color is most often specified by:

- a valid color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

Other Elements

Setting the background color for any HTML elements:

Example

Here, the `<h1>`, `<p>`, and `<div>` elements will have different background colors:

```
h1 {  
  background-color: green;  
}  
  
div {  
  background-color: lightblue;  
}  
  
p {  
  background-color: yellow;  
}
```

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
  h1 {  
    background-color: green;  
  }  
  
  div {  
    background-color: lightblue;  
  }  
  
  p {  
    background-color: yellow;  
  }  
</style>  
</head>  
<body>  
  
  <h1>CSS background-color example!</h1>  
  <div>  
    This is a text inside a div element.  
    <p>This paragraph has its own background color.</p>  
    We are still in the div element.  
  </div>  
  
</body>  
</html>
```

CSS Backgrounds

Opacity / Transparency

The `opacity` property specifies the opacity/transparency of an element. It can take a value from 0.0 - 1.0. The lower value, the more transparent:

Example

```
div {  
  background-color: green;  
  opacity: 0.3;  
}
```

```
<!DOCTYPE html>  
<html>  
  <head>  
    <style>  
      div { background-color: green;  
  
      div.first { opacity: 0.1;}  
  
      div.second { opacity: 0.3;}  
  
      div.third { opacity: 0.6;}  
    </style>  
  </head>  
  <body>  
  
    <h1>Transparent Box</h1>  
    <p>  
      When using the opacity property to add transparency to the background of an element,  
      all of its child elements become transparent as well.  
      This can make the text inside a fully transparent element hard to read:</p>  
  
    <div class="first">  
      <h1>opacity 0.1</h1>  
    </div>  
  
    <div class="second">  
      <h1>opacity 0.3</h1>  
    </div>  
  
    <div class="third">  
      <h1>opacity 0.6</h1>  
    </div>  
  
    <div>  
      <h1>opacity 1 (default)</h1>  
    </div>  
  
  </body>  
</html>
```

Note: When using the `opacity` property to add transparency to the background of an element, all of its child elements inherit the same transparency. This can make the text inside a fully transparent element hard to read.

CSS Backgrounds

Transparency using RGBA

If you do not want to apply opacity to child elements, like in our example above, use **RGBA** color values. The following example sets the opacity for the background color and not the text:



An RGBA color value is specified with: `rgba(red, green, blue, alpha)`. The *alpha* parameter is a number between 0.0 (fully transparent) and 1.0 (fully opaque).

```
<!DOCTYPE html>
<html>
<head>
<style>
div { background: rgb(0, 128, 0);}
div.first { background: rgba(0, 128, 0, 0.1);}
div.second { background: rgba(0, 128, 0, 0.3);}
div.third { background: rgba(0, 128, 0, 0.6);}
</style>
</head>
<body>
<h1>Transparent Box</h1>
<p>With opacity:</p>
<div style="opacity:0.1;"> <h1>10% opacity</h1> </div>
<div style="opacity:0.3;"> <h1>30% opacity</h1> </div>
<div style="opacity:0.6;"> <h1>60% opacity</h1> </div>
<div> <h1>opacity 1</h1> </div>

<p>With RGBA color values:</p>
<div class="first"> <h1>10% opacity</h1> </div>
<div class="second"> <h1>30% opacity</h1> </div>
<div class="third"> <h1>60% opacity</h1> </div>
<div> <h1>default</h1> </div>
<p>Notice how the text gets transparent as well as the background color when using the
opacity property.</p>
</body>
</html>
```

CSS Backgrounds

CSS background-image

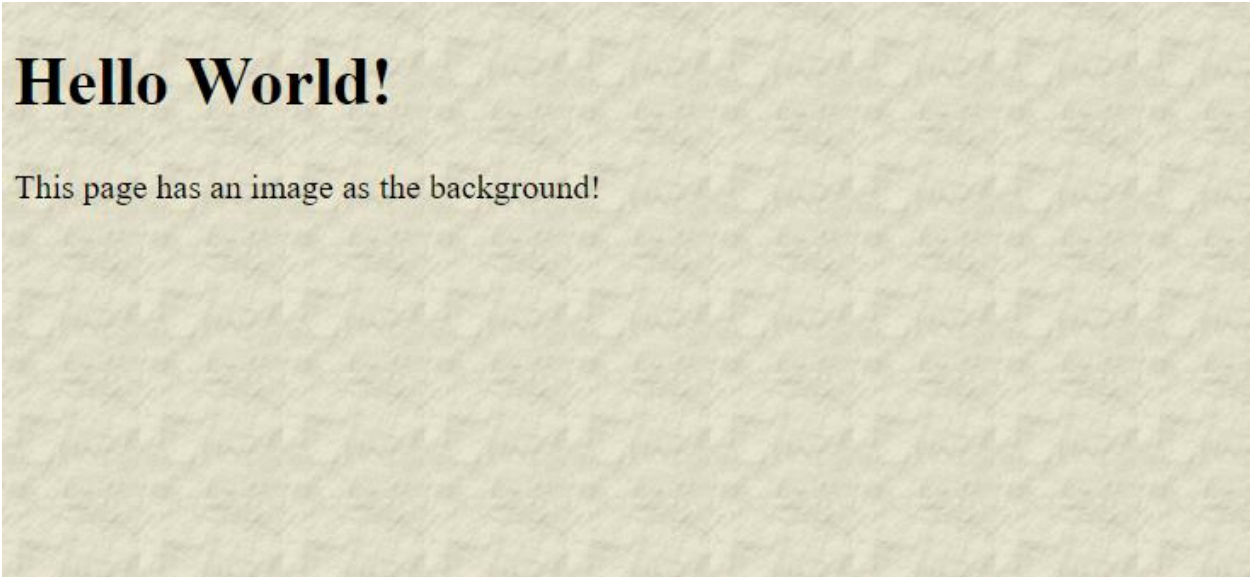
The `background-image` property specifies an image to use as the background of an element.

By default, the image is repeated so it covers the entire element.

Example

Set the background image for a page:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-image: url("paper.gif");
}
</style>
</head>
<body>
<h1>Hello World!</h1>
<p>This page has an image as the background!</p>
</body>
</html>
```



Hello World!

This page has an image as the background!

CSS Backgrounds

Example

This example shows a **bad combination** of text and background image. The text is hardly readable:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-image: url("bgdesert.jpg");
}
</style>
</head>
<body>

<h1>Hello World!</h1>
<p>This text is not easy to read on this background image.</p>

</body>
</html>
```



Note: When using a background image, use an image that does not disturb the text.

The background image can also be set for specific elements, like the <p> element:

Example

```
p {
  background-image: url("paper.gif");
}
```

CSS Backgrounds

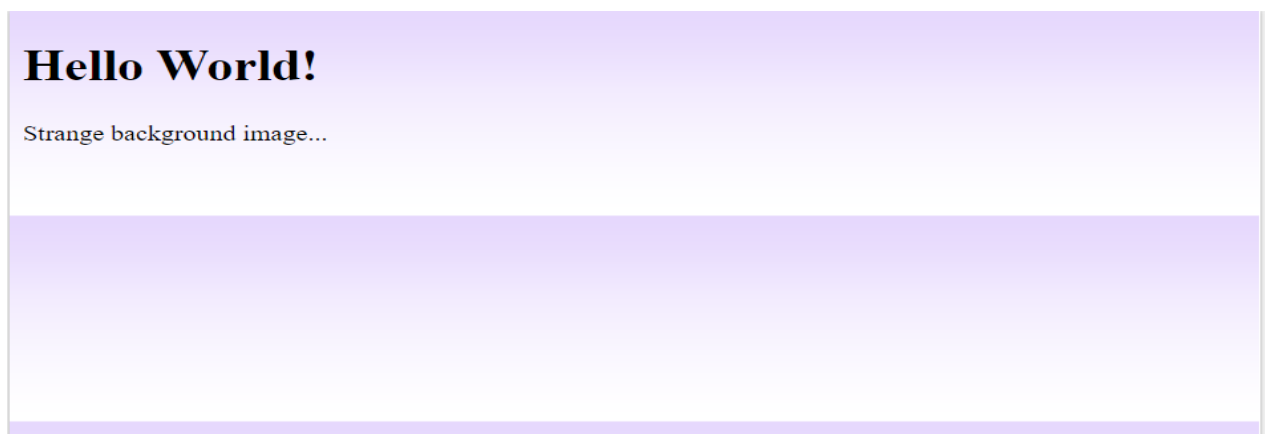
CSS background-repeat

By default, the `background-image` property repeats an image both horizontally and vertically.

Some images should be repeated only horizontally or vertically, or they will look strange, like this:

Example

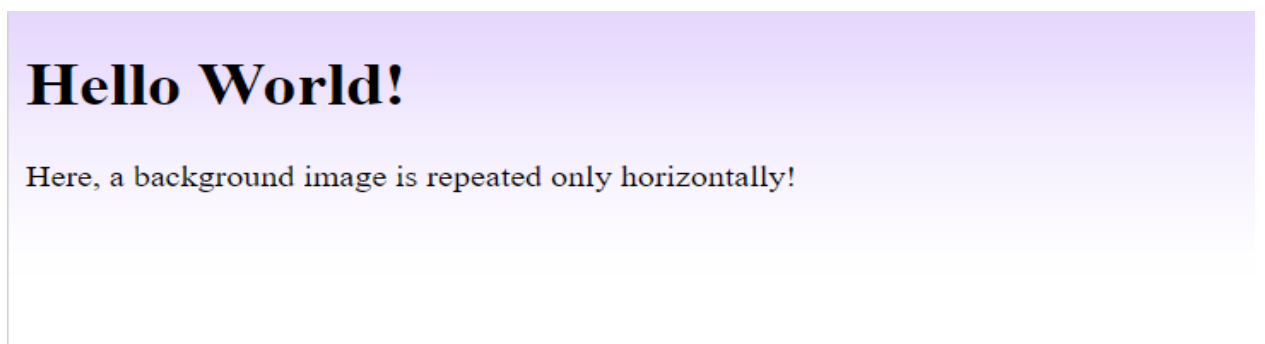
```
body {  
  background-image: url("gradient_bg.png");  
}
```



If the image above is repeated only horizontally (`background-repeat: repeat-x;`), the background will look better:

Example

```
body {  
  background-image: url("gradient_bg.png");  
  background-repeat: repeat-x;  
}
```



Tip: To repeat an image vertically, set `background-repeat: repeat-y;`

CSS Backgrounds

CSS background-repeat: no-repeat

Showing the background image only once is also specified by the `background-repeat` property:

Example

Show the background image only once:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
}
```

Hello World!

W3Schools background image example.

The background image is only showing once, but it is disturbing the reader!



In the example above, the background image is placed in the same place as the text. It can change the position of the image, so that it does not disturb the text too much.

CSS Backgrounds

CSS background-position

The `background-position` property is used to specify the position of the background image.

Example

Position the background image in the top-right corner:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
}
```

Hello World!

W3Schools background no-repeat, set position example.

Now the background image is only shown once, and positioned away from the text.

In this example we have also added a margin on the right side, so the background image will never disturb the text.



CSS Backgrounds

CSS background-attachment

The `background-attachment` property specifies whether the background image should scroll or be fixed (will not scroll with the rest of the page):

Example

Specify that the background image should be fixed:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
  background-attachment: fixed;  
}
```

Example

Specify that the background image should scroll with the rest of the page:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
  background-attachment: scroll;  
}
```

CSS Background Shorthand

To shorten the code, it is also possible to specify all the background properties in one single property. This is called a shorthand property.

Instead of writing:

```
body {  
  background-color: #ffffff;  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
}
```

It can use the shorthand property `background`:

```
body {  
  background: #ffffff url("img_tree.png") no-repeat right top;  
}
```

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When using the shorthand property the order of the property values is:

- `background-color`
- `background-image`
- `background-repeat`
- `background-attachment`
- `background-position`

It does not matter if one of the property values is missing, as long as the other ones are in this order. Note that we do not use the `background-attachment` property in the examples above, as it does not have a value.

All CSS Background Properties

Property	Description
<u>background</u>	Sets all the background properties in one declaration
<u>background-attachment</u>	Sets whether a background image is fixed or scrolls with the rest of the page
<u>background-clip</u>	Specifies the painting area of the background
<u>background-color</u>	Sets the background color of an element
<u>background-image</u>	Sets the background image for an element
<u>background-origin</u>	Specifies where the background image(s) is/are positioned
<u>background-position</u>	Sets the starting position of a background image
<u>background-repeat</u>	Sets how a background image will be repeated
<u>background-size</u>	Specifies the size of the background image(s)